

Tooling.

- Tools: Python 3, NumPy, pandas, SciPy, matplotlib.
- Numbers: generated from saved result files, none typed by hand.
- Code: `hw4/code/` (`run_all.py` runs every step). Code and write-up drafted with Claude Code.
- Significance level for every test: $\alpha = 0.05$, fixed before the tests were run.

1 Problem 1: Two binary genomic variants

Data. `p1a.csv` holds 199 plants (rows) and two genomic variants (columns). A 1 means the variant is present. The file `p1b.csv` (199×15) also came with the data, but the assignment does not use it.

1.1 Contingency table

Table 1: Observed counts of the two variants, with expected counts under independence in parentheses. Source: `hw4/results/p1_contingency.csv`.

	Variant 2 = 0	Variant 2 = 1	Row total
Variant 1 = 0	128 (133.28)	21 (15.72)	149
Variant 1 = 1	50 (44.72)	0 (5.28)	50
Column total	178	21	199

Expected count: $E_{ij} = (\text{row}_i \times \text{col}_j)/n$. The key cell is $n_{11} = 0$: no plant carries both variants, while independence predicts 5.28.

1.2 Statistics

Notation: n_{11} = both present, n_{10} = only variant 1, n_{01} = only variant 2.

- **Mutual information** (log base 2, so the unit is bits, lecture 15): $MI(X, Y) = H(X) + H(Y) - H(X, Y) = \sum_{x,y} p(x, y) \log_2 \frac{p(x,y)}{p(x)p(y)}$.
- **Jaccard index**: $J = n_{11}/(n_{11} + n_{10} + n_{01})$.
- **Chi-squared**: $\chi^2 = \sum_{i,j} (O_{ij} - E_{ij})^2 / E_{ij}$, with 1 degree of freedom.

Table 2: Association statistics for `p1a.csv`. Source: `hw4/results/p1_stats.csv`.

Statistic	Value	Significance
Entropy $H(\text{variant 1}), H(\text{variant 2})$	0.8133, 0.4863 bits	
Mutual information MI	0.0470 bits (0.0326 nats)	permutation $p = 0.0018$
Normalized MI, $MI/\sqrt{H(X)H(Y)}$	0.0747	
Jaccard index J	0.0000	
Chi-squared χ^2 (no continuity correction)	7.8784	$p = 0.0050$
Chi-squared with Yates correction	6.4560	$p = 0.0111$

MI significance. I shuffled variant 2 10,000 times and recomputed MI each time (lecture 15, slide 26). The p-value is the share of shuffles with MI at least as large as the observed value.

Findings.

- The two variants are associated. The chi-squared test rejects independence at $\alpha = 0.05$ ($p = 0.0050$). The Yates-corrected test agrees ($p = 0.0111$). So does the MI permutation test ($p = 0.0018$).
- The association is negative: the two variants never occur together. The table shows 0 plants with both, against 5.28 expected.
- This explains the Jaccard index of 0.0000. Jaccard counts only shared presence (n_{11}), so it is 0 here. Jaccard shows that the variants do not co-occur, but it cannot tell “unrelated” from “mutually exclusive”.
- MI is small in absolute terms (0.0470 bits, normalized 0.0747) because variant 2 is rare (21 of 199 plants). Knowing one variant removes little of the other’s uncertainty. MI has no direction, so it does not show that the association is negative.
- The smallest expected count is 5.28 (≥ 5), so the chi-squared approximation is acceptable.

2 Problem 2: Pairs of continuous gene expressions

Method. For each file I computed the Pearson correlation r between gene 1 and gene 2. The null hypothesis is no association, $H_0 : \rho = 0$. The test statistic is $t = r\sqrt{n-2}/\sqrt{1-r^2}$ with $n-2$ degrees of freedom (lecture 14, slide 34). The p-value is two-sided. I reject H_0 when $p < \alpha = 0.05$.

Table 3: Pearson correlation and p-value for each pair. Source: `hw4/results/p2_stats.csv`.

Part	File	n	r	t	p	H_0 at $\alpha = 0.05$
(a)	<code>p2a.csv</code>	2,400	0.3809	20.17	1.04×10^{-83}	reject
(b)	<code>p2b.csv</code>	110	0.9312	26.55	3.74×10^{-49}	reject
(c)	<code>p2c.csv</code>	2,100	0.0412	1.89	0.0592	do not reject

2.1 Part (a): `p2a.csv`

The Pearson correlation is $r_a = 0.3809$ with $p_a = 1.04 \times 10^{-83}$ ($n = 2,400$). Because $p_a < 0.05$, I reject the null hypothesis. There is a statistically significant association between the two genes. The direction is positive: when gene 1 is higher, gene 2 tends to be higher. The magnitude is weak to moderate: the points form a wide cloud around an upward trend (Figure 1a).

2.2 Part (b): p2b.csv

The Pearson correlation is $r_b = 0.9312$ with $p_b = 3.74 \times 10^{-49}$ ($n = 110$). Because $p_b < 0.05$, I reject the null hypothesis. There is a statistically significant association between the two genes. The direction is positive, and the magnitude is very strong: the points lie close to a straight line (Figure 1b).

2.3 Part (c): p2c.csv

The Pearson correlation is $r_c = 0.0412$ with $p_c = 0.0592$ ($n = 2,100$). Because $p_c > 0.05$, I do not reject the null hypothesis. By the Pearson test, there is no statistically significant linear association. The estimated linear association is close to zero, with a slightly positive sign. However, the scatter plot shows a strong non-linear relationship (Figure 1c), discussed in part (e).

2.4 Part (d): Scatter plots

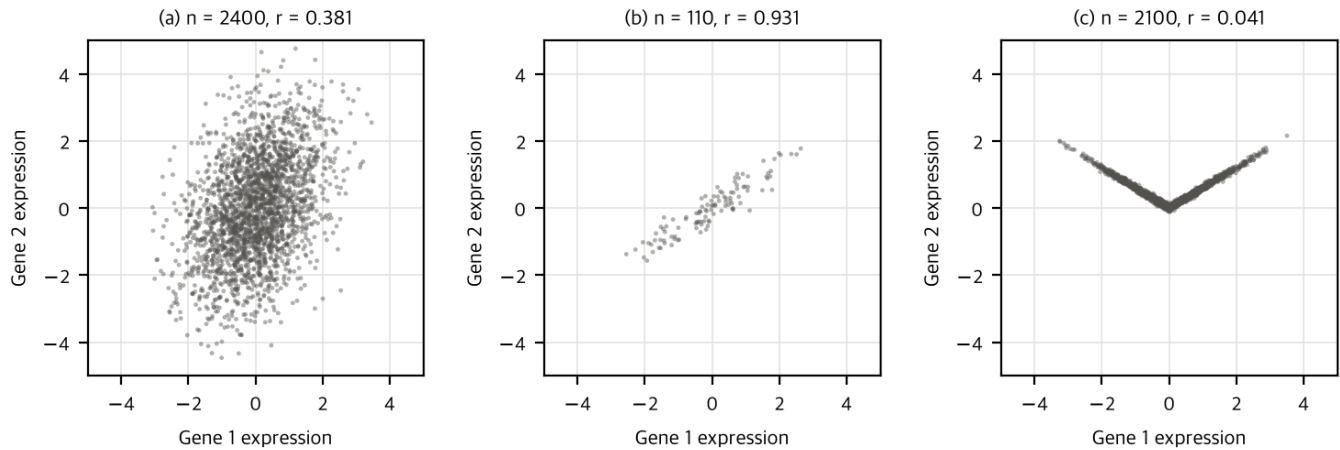


Figure 1: Gene 1 (x) against gene 2 (y) for each file, on the same axis range. Panel titles give n and Pearson r . Source: hw4/figures/.

2.5 Part (e): Discussion

Table 4: Ranking of the three pairs, strongest association first.

Criterion	Ranking	Basis
Correlation r	(b) > (a) > (c)	$0.9312 > 0.3809 > 0.0412$
p-value (smaller = stronger evidence)	(a) > (b) > (c)	$1.04 \times 10^{-83} < 3.74 \times 10^{-49} < 0.0592$
Scatter plots (my judgment)	(c) > (b) > (a)	shape and spread of the points

1. **By correlation:** (b), then (a), then (c). Pair (b) is almost perfectly linear ($r = 0.9312$). Pair (a) is weakly linear ($r = 0.3809$). Pair (c) has r near zero (0.0412).
2. **By p-value:** (a), then (b), then (c). Pair (a) has the smallest p-value (1.04×10^{-83}), even though its correlation is much weaker than that of (b). Pair (c) is not significant ($p = 0.0592$).
3. **By the scatter plots:** (c), then (b), then (a). In (c) gene 2 is almost exactly a V-shaped function of gene 1, close to $|\text{gene 1}|$. Given gene 1, gene 2 is nearly fixed, so the dependence is the strongest of the

three, but it is not linear. In (b) the points lie tightly along a straight line. In (a) the points form a broad tilted cloud.

4. **Do they agree?** No. Each criterion picks a different strongest pair: (b) by r , (a) by p , and (c) by eye. Two reasons explain this:

- **The p-value depends on sample size, not only on strength.** It measures the evidence against $\rho = 0$, not the size of the association. Pair (a) has $n = 2,400$ and pair (b) has $n = 110$. With 2,400 samples, even a modest $r = 0.3809$ gives a large $t = 20.17$. Pair (b) has a much larger r but only 110 samples, so $t = 26.55$ is only slightly larger. Because the degrees of freedom are much smaller for (b), its p-value ends up larger than that of (a). Both are far below α . The difference between them reflects n , not strength.
- **Pearson's r measures only linear association.** In (c) the left half of the points slopes down and the right half slopes up. The two halves cancel, so $r \approx 0$ and the test does not reject H_0 . The genes are still strongly dependent: the correlation of |gene 1| with gene 2 is 0.9911. Spearman's rank correlation (0.0338) also misses it, because the relation is not monotonic. A measure that handles non-linear dependence, such as mutual information or MIC (lecture 15), would rank (c) high.

So a large r shows a strong linear relation, and a small p shows that an effect is unlikely to be zero. Neither replaces looking at the data.

Side note. The gene 1 values of `p2b.csv` equal the first 110 gene 1 values of `p2c.csv` (checked: yes). The two files share one variable but differ in gene 2 and in the shape of the relation.